

Drawbacks of Matrix Factorization Based Method on Categorical Data

	Income	Credit	Purchase
John	High	High	High
Amy	High	High	High
Carl	Low	Low	Low
Mary	Low	Low	Low
Tom	High	Low	Medium
Jim	High	Low	Low

When a categorical attribute has many different values, the binary data matrix B would have many columns since we need to create a separate column (i.e., binary feature) for each categorical attribute value. This might make the matrix factorization process computationally intensive and the detection results hard to interpret

	I/H	I/M	I/L	C/H	C/M	C/L	P/H	P/M	P/L
John	1	0	0	1	0	0	1	0	0
Amy	1	0	0	1	0	0	1	0	0
Carl	0	0	1	0	0	1	0	0	1
Mary	0	0	1	0	0	1	0	0	1
Tom	1	0	0	0	0	1	0	1	0
Jim	1	0	0	0	0	1	0	0	1

(a) Representing the input categorical data as binary matrix B

1	0
1	0
0	1
0	1
0.5	0.5
0.3	0.7

 $\times$

1	0	0	1	0	0	1	0	0
0	0	1	0	0	1	0	0	1

Matrix G

Matrix F

(b) The two factorized matrices

	I/H	I/M	I/L	C/H	C/M	C/L	P/H	P/M	P/L
John	1	0	0	1	0	0	1	0	0
Amy	1	0	0	1	0	0	1	0	0
Carl	0	0	1	0	0	1	0	0	1
Mary	0	0	1	0	0	1	0	0	1
Tom	0.5	0	0.5	0.5	0	0.5	0.5	0	0.5
Jim	0.3	0	0.7	0.3	0	0.7	0.3	0	0.7

(c) The reconstructed matrix